Math 231A HW 6

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Problem 1

Leg $\mathfrak{g}$ be a Lie algebra, and $(\_,\_) : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ a symmetric ad-invariant bilinear form on $\mathfrak{g}$ (i.e. for any $a, x, y \in \mathfrak{g}$, we have $(x, y) = (y, x)$ and $([a, x], y) + (x, [a, y]) = 0$, which is the same as $([a, x], y) = -(x, [a, y])$). Show that the element $w \in (\mathfrak{g}^*)^{\otimes 3}$ given by

$$w(x, y, z) = ([x, y], z) \tag{1.1}$$

is skew-symmetric and ad-invariatnt.

Solution:

I. Skew-Symmetric:

It suffices to verify that if two entries are the same, then it evaluates to be 0. Given any $x, y \in \mathfrak{g}$, it satisfies the following three relations:

$$w(x, x, y) = ([x, x], y) = (0, y) = 0 \tag{1.2}$$

$$w(x, y, x) = ([x, y], x) = -(y, [x, x]) = 0 \tag{1.3}$$

$$w(y, x, x) = ([y, x], x) = (-[x, y], x) = -(y, [x, x]) = -(y, 0) = 0 \tag{1.4}$$

This shows that w is skew-symmetric.

II. Ad-invariance:

The goal is to prove that for any $a, x, y, z \in \mathfrak{g}$, the following relation holds:

$$w([a, x], y, z) + w(x, [a, y], z) + w(x, y, [a, z]) = 0 \tag{1.5}$$

Which, based on the given relation between w and $(\_,\_)$, together with Jacobi's Identity, we get:

$$[[a, x], y] = -[[x, y], a] - [[y, a], x] \tag{1.6}$$

$$\begin{aligned} w([a, x], y, z) &= ([[a, x], y], z) = -([[x, y], a], z) - ([[y, a], x], z) \\ &= ([a, [x, y]], z) + ([[a, y], x], z) = -([x, y], [a, z]) + w([a, y], x, z) \\ &= -w(x, y, [a, z]) - w(x, [a, y], z) \end{aligned} \tag{1.7}$$

Hence, we get that $w([a, x], y, z) + w(x, [a, y], z) + w(x, y, [a, z]) = 0$.

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Problem 2

Let C be the standard cube in $\mathbb{R}^3 : C := \{|x_i| \leq 1\}$, and let S be the set of faces of C (thus, S consists of 6 elements). Consider the 6-dimensional complex vector V space of functions on S, and define $A : V \to V$ by

$$(Af)(\sigma) = \frac{1}{4} \sum_{\sigma'} f(\sigma') \tag{2.1}$$

where the sum is taken over all faces σ' which are neighbors of σ (i.e. have a common edge with σ). The goal of this problem is to diagonalize A.

(1) Let $G = \{g \in O(3, \mathbb{R}) \mid g(C) = C\}$ be the group of symmetries of C. Show that A commutes with the natural action of G on V.

(2) Let $z = -I \in G$. Show that as a representation of G, V can be decomposed in the direct sum

$$V = V_+ \oplus V_-, \quad V_\pm = \{f \in V \mid zf = \pm f\} \tag{2.2}$$

(3) Show that as a representation of G, V_+ can be decomposed in the direct sum

$$V_+ = V_+^0 \oplus V_+^1, \quad V_+^0 := \left\{ f \in V_+ \mid \sum_\sigma f(\sigma) = 0 \right\}, \quad V_+^1 = \mathbb{C} \cdot 1 \tag{2.3}$$

where 1 denotes the constant function on S whose value at every $\sigma \in S$ is 1.

(4) Find the eigenvalues of A on V_-, V_+^0, V_+^1.

[Note: in fact, each of V_-, V_+^0, V_+^1 is an irreducible representation of G, but you do not need this fact].

Solution:

(1) Since G preserves the whole cube C by „rotating“ and/or „reflecting“ the cube in ways that send faces to faces, in particular it sends adjacent faces to adjacent faces. Hence, for any face $\sigma \in S$, if $\sigma' \in S$ is one of its adjacent faces, for any $g \in G$ one still has $g \cdot \sigma$ and $g \cdot \sigma'$ being adjacent faces (and this is an if and only if relation, because g is invertible).

Hence, one can define an action $G \curvearrowright V$ as follow:

$$\forall f \in V, \quad \forall g \in G, \quad \forall \sigma \in S, \qquad (g \cdot f)(\sigma) := f(g \cdot \sigma) \tag{2.4}$$

Now, notice that for all $f \in V$ and any $g \in G$, given any $\sigma \in S$, we have the following:

$$g \cdot f(\sigma) = f(g \cdot \sigma), \quad A(g \cdot f)(\sigma) = \sum_{\sigma'} (g \cdot f)(\sigma') = \sum_{\sigma'} f(g \cdot \sigma') \tag{2.5}$$

Which, notice that the collection σ' are the adjacent faces of σ, hence $g \cdot \sigma'$ are the adjacent faces of $g \cdot \sigma$. Therefore, we also have the following relation:

$$(g \cdot Af)(\sigma) = (Af)(g \cdot \sigma) = \sum_{g \cdot \sigma'} f(g \cdot \sigma') = A(g \cdot f)(\sigma) \tag{2.6}$$

Since $\sigma \in S$ is arbitrary, this proves that $A(g \cdot f) = g \cdot (Af)$ as functions, hence as a linear operator on V, A is commuting with the action of G on V.

(2) Given $z = -I \in G$, for any face $\sigma \in S$, let $-\sigma \in S$ denotes the face of C on the opposite side of σ. Then, notice that $-\sigma = z(\sigma)$ (since $-\sigma$ precisely contains all negative vectors of σ). Hence, if consider any funciton $f \in V_{\pm}$, since it satisfies $z \cdot f = \pm f$, then we must have:

$$\pm f(\sigma) = (z \cdot f)(\sigma) = f(z \cdot \sigma) = f(-\sigma) \tag{2.7}$$

Hence, for V_+, we have $f(-\sigma) = f(\sigma)$ (indicating opposite faces have the same value), while for V_-, we have $f(-\sigma) = -f(\sigma)$ (indicating opposite faces have neative values).

As a result, because for the G action on S, any $g \in G$ and $\sigma \in S$ has $-(g \cdot \sigma) = g \cdot (-\sigma)$ (i.e. the G-action on S preserves opposite faces), then given any $f \in V_{\pm}$, we have the following for all $\sigma \in S$:

$$(z \cdot (g \cdot f))(\sigma) = (g \cdot f)(-\sigma) = f(g \cdot (-\sigma)) = f(-(g \cdot \sigma)) = \pm f(g \cdot \sigma) = \pm (g \cdot f)(\sigma) \tag{2.8}$$

where the second last equality is coming from the fact that $\pm f(\sigma') = (z \cdot f)(\sigma') = f(-\sigma')$ (where $\pm$ depends on $V_{\pm}$). So, since the above indicates that $z \cdot (g \cdot f) = \pm (g \cdot f)$ for all $g \in G$ and $f \in V_{\pm}$, showing that $g \cdot f \in V_{\pm}$ also, hence $V_{\pm}$ each form a subrepresentation of G.

To prove that $V_{\pm}$ sums to V, first for any function $f \in V$, consider the pair $\frac{f + z \cdot f}{2}$ and $\frac{f - z \cdot f}{2}$, they satisfy the following:

$$z \cdot \left(\frac{f + z \cdot f}{2}\right) = \frac{z \cdot f + z^2 \cdot f}{2} = \frac{z \cdot f + f}{2} = \frac{f + z \cdot f}{2} \tag{2.9}$$

$$z \cdot \left(\frac{f - z \cdot f}{2}\right) = \frac{z \cdot f - z^2 \cdot f}{2} = \frac{z \cdot f - f}{2} = -\frac{f - z \cdot f}{2} \tag{2.10}$$

This indicates that $\frac{f + z \cdot f}{2} \in V_+$, while $\frac{f - z \cdot f}{2} \in V_-$, and $\frac{f + z \cdot f}{2} + \frac{f - z \cdot f}{2} = f \in V$. Hence, this implies $V = V_+ + V_-$ as vector spaces.

Finally, the reason why it's a direct sum, is because for any $f \in V_+ \cap V_-$, we must have $z \cdot f = f$ and $z \cdot f = -f$ simultaneously, so $f = -f$, or $f = 0$. Hence, with intersection being trivial, $V_+ \oplus V_- = V$.

(3) First to prove that $V_+ = V_+^0 + V_+^1$, recall that V_+ is the space of function where every two opposite faces yield the same value for every funciton in V_+. Hence, there are total of 3 faces (or 3 directions) that may consist of different function values (WLOG, say in $\mathbb{R}^3$ these three faces are normal to e_1, e_2, e_3 three vectors). Then, consider the permutation matrices $g = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, g^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in G$ (which they preserve the cube, since it's only swapping the 3 standard basis vectors). Then, since for any function $f \in V_+$, one has $\frac{f + g \cdot f + g^2 \cdot f}{3}$ that yields the same value for every face (for definiteness, say pick face at direction e_1, then $g \cdot f(e_1) = f(g \cdot e_1) = f(e_3)$, and $g^2 \cdot f(e_1) = f(g^2 \cdot e_1) = f(e_2)$, so $\frac{f + g \cdot f + g^2 \cdot f}{3}(e_1) = \frac{f(e_1) + f(e_2) + f(e_3)}{3}$; with $f \in V_+$, then the same statement is true for $\pm e_1$, and is true also for $\pm e_2, \pm e_3$). Hence, $\frac{f + g \cdot f + g^2 \cdot f}{3} \in V_+^1 = \mathbb{C} \cdot 1$.

Also, notice that the sum $\sum_\sigma f(\sigma) = 2(f(e_1) + f(e_2) + f(e_3))$ (since the other three faces ar the opposite of these three, which satisfies $f(-e_i) = f(e_i)$, since $f \in V_+$), while $\sum_\sigma \frac{f + g \cdot f + g^2 \cdot f}{3}(\sigma) = 6 \cdot \frac{f(e_1) + f(e_2) + f(e_3)}{3} = 2(f(e_1) + f(e_2) + f(e_3))$. Hence, if consider the function $h = f - \frac{f + g \cdot f + g^2 \cdot f}{3}$, it satisfies $\sum_\sigma h(\sigma) = 0$, hence $h \in V_+^0$. And, notice that $h + \frac{f + g \cdot f + g^2 \cdot f}{3} = f$, which also shows that $V_+ = V_+^0 + V_+^1$.

Then, to show it's a direct sum, it's because for any $f \in V_+^0 \cap V_+^1$, we have $f = k \cdot 1$ for some $k \in \mathbb{C}$; also, we have $\sum_\sigma f(\sigma) = \sum_\sigma k = 6k = 0$ based on the property of V_+^0, hence we must have $k = 0$, showing $f = 0$. So, $V_+^0 \oplus V_+^1 = V_+$.

Finally, the two are subrepresentations, simply just because the G-action on C sends faces to faces, so for any $f \in V_+^0$, since $\sum_\sigma f(\sigma) = 0$, then for any $g \in G$, $\sum_\sigma (g \cdot f)(\sigma) = \sum_\sigma f(g \cdot \sigma) = 0$ (since g just permutes S), showing that $g \cdot f \in V_+^0$; also, for any $h \in V_+^1$ (where $h = k \cdot 1$ for some $k \in \mathbb{C}$), then $g \cdot h(\sigma) = h(g \cdot \sigma) = k$ for any $\sigma \in S$, showing $g \cdot h \in V_+^1$. Hence, V_+^0, V_+^1 are both invariant under G, which are subrepresentations of G.

(4) First, for any $f \in V_-$, since it satisfies $f(-\sigma) = -f(\sigma)$ for all face $\sigma \in S$, then it satisfies $(Af)(\sigma) = \sum_{\sigma'} f(\sigma') = 0$ (since let σ_1, σ_2 be two adjacent sides of faces that're not opposite face of each other, the other two adjacent sides are given by $-\sigma_1, -\sigma_2$; so, the sum becomes $f(\sigma_1) + f(\sigma_2) + f(-\sigma_1) + f(-\sigma_2) = 0$ based on the fact that $f(-\sigma) = -f(\sigma)$). Hence, this shows that $Af = 0$ (given $f \in V_-$), so A has eigenvalue 0 on V_- (and whole V_- is contained in the eigenspace of 0).

Then, for any $f \in V_+^0 \subseteq V_+$, since every face $\sigma \in S$ satisfies $f(-\sigma) = f(\sigma)$ (the property of V_+), and also $\sum_\sigma f(\sigma) = 0$, then since the adjacent faces of σ precisely excluded σ and $-\sigma$, then if the collection of σ' runs through the adjacent faces of σ, it satisfies $\sum_{\sigma'} f(\sigma') + f(\sigma) + f(-\sigma) = \sum_{\sigma'' \in S} f(\sigma'') = 0$, hence $\sum_{\sigma'} f(\sigma') = -f(\sigma) - f(-\sigma) = -2f(\sigma)$. So, we have that $(Af)(\sigma) = \frac{1}{4}\sum_{\sigma'} f(\sigma') = -\frac{1}{2} f(\sigma)$, showing that $Af = -\frac{1}{2} f$ (given $f \in V_+^0$), so A has eigenvalue $-\frac{1}{2}$ on V_+^0 (also, all V_+^0 is contained in the eigenspace of $-\frac{1}{2}$).

Finally, for any $f \in V_+^1$, since $f = k \cdot 1$ for some $k \in \mathbb{C}$, then for all $\sigma \in S$, it satisfies $f(\sigma) = k$. Hence $(Af)(\sigma) = \frac{1}{4}\sum_{\sigma'} f(\sigma') = \frac{1}{4} \cdot 4k = k = f(\sigma)$, showing that $Af = f$ (given $f \in V_+^1$). Hence, A has eigenvalue 1 on V_+^1 (again, all V_+^1 is contained in the eigenspace of 1).

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Problem 3

Show that if V is finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$, then $V \cong \bigoplus n_k V_k$ (or $\bigoplus V_k^{n_k}$), and $n_k = \dim V[k] - \dim V[k+2]$ (where $V[k]$ representes the eigenspace of k as an eigenvalue of $h \in \mathfrak{sl}(2, \mathbb{C})$). Show also that $\sum n_{2k} = \dim V[0]$, $\sum n_{2k+1} = \dim V[1]$.

Solution: Since V is a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$-Representation, then it's completely reducible, and each irreducible component must be isomorphic to some V_k. Hence, $V \cong \bigoplus V_k^{n_k}$ for finitely many nonnegative integer k (where n_k represents the number of copies of V_k, so only for finite k we have $n_k > 0$).

Now, fix a specific nonnegative integer k. For any $l \in \mathbb{Z}$, the Representation V_{k+l} has basis elements $x^p y^q$, where $p + q = k + l$, and it satisfies $h \cdot (x^p y^q) = (p - q) x^p y^q$. Hence, each unique pair (p, q) corresponds to a unique eigenvalue of h on V_{k+l}, and since this list is a basis of V_{k+2l}, it implies that h has $(k + l) + 1 = \dim(V_{k+l})$ distinct eigevalues on V_{k+l}, hence each eigenvalue corresponds to a 1-dimensional eigenspace. So, if k is an eigenvalue of h on V_{k+l}, it has at most 1-dimensional eigenspace.

Also, if k is an eigenvalue of h on V_{k+l}, it must be one of th eigenvalues of the form $(p - q)$, where $p + q = k + l$. Hence, for some p, q satisfying $p + q = k + l$, one has $k = p - q$, showing that $p + q = k + l = p - q + l$, so $l = 2q$ (where q is a nonnegative integer). Hence, every V_{k+l} which has k as an eigenvalue of h must be in the form V_{k+2q} for some nonnegative integer q. Moreover, this is an if and only if, since conversely for any nonnegative integer q, the polynomial $x^{k+q}y^q \in V_{k+2q}$ satisfies $h \cdot (x^{k+q}x^q) = ((k+q)-q)x^{k+q}y^q = k \cdot x^{k+q}y^q$, showing h as an operator on V_{k+2q} has eigenvalue k).

Hence, we concluded that k is an eigenvalue of V_{k+l} iff $l = 2q$ for some nonnegative integer q, and it corresopnds to a 1-dimensional eigenspace in V_{k+l}. Hence, given that $V \cong \bigoplus V_{k'}^{n_{k'}}$ (where for finite amount of k' we have $n_{k'} > 0$), for each fixed k, the only $k' \in \mathbb{N}$ that has $V_{k'}$ contains a nontrivial eigenvector corresponding to eigenvalue k, are $k' = k + 2q$ for $q \in \mathbb{N}$.

Also, for every $v \in V[k]$, since it can be uniquely expressed as finite sums of vectors in each $V_{k'}$ (based on the direct sum argument), while each $V_{k'}$ is invariant under h, then if $v = v_{k_1'} + ... + v_{k_m'}$ (where each nonzero $v_{k_i'} \in V_{k_i'}$), we have $kv_{k_1'} + ... + kv_{k_m'} = kv = h \cdot v = h \cdot v_{k_1'} + ... + h \cdot v_{k_m'}$, and each $kv_{k_i'}, h \cdot v_{k_i'} \in V_{k_i'}$. Hence, by uniqueness $h \cdot v_{k_i'} = kv_{k_i'}$, showing each $V_{k_i'}$ contains some eigenvector corresponding to eigenvalue k, hence $k_i' = k + 2q_i$ for some $q_i \in \mathbb{N}$.

This shows that $V[k]$ is in fact a direct sum of eigenspace of k in each V_{k+2q}, denote as $E_{k+2q}(k) \subset V_{k+2q}$. Since each $E_{k+2q}(k)$ is 1-dimensional, then $\dim V[k] = \sum_{q\geq 0} n_{k+2q}$ (since each $q \geq 0$ has V_{k+2q} containing a copy of $E_{k+2q}(k)$, while there are n_{k+2q} copies of V_{k+2q} in V). As a result, we have the following for n_k:

$$n_k = \sum_{q\geq 0} n_{k+2q} - \sum_{q\geq 1} n_{k+2q} = \dim V[k] - \sum_{q\geq 0} n_{(k+2)+2q} = \dim V[k] - \dim V[k+2] \tag{3.1}$$

Finally, with the formula derived above about n_k, using Telescoping series, we have the following:

$$\sum_{k=0}^{l} n_{2k} = \sum_{k=0}^{l} (\dim V[2k] - \dim V[2k+2]) = \dim V[0] - \dim V[2l+2] \tag{3.2}$$

$$\sum_{k=0}^{l} n_{2k+1} = \sum_{k=0}^{l} (\dim V[2k+1] - \dim V[2k+3]) = \dim V[1] - \dim V[2l+3] \tag{3.3}$$

Since $V \cong \bigoplus V_{k'}^{n_{k'}}$ is finite-dimensional, then there exists some $m \in \mathbb{N}$, such that $k' \geq m$ implies $n_{k'} = 0$. Then, since k' can only be an eigenvalue of h in V iff there is a nontrivial copy of $V_{k'+2q}$ for some $q \in \mathbb{N}$, then with $k' \geq m$, k' is not an eigenvalue of h over V (since there's no nontrivial copy of $V_{k'+2q}$ for any $q \in \mathbb{N}$).

Hence, for any $l \geq m$, we have $\sum_{k=0}^{l} n_{2k} = \dim V[0] - \dim V[2l+2] = \dim V[0]$, and $\sum_{k=0}^{l} n_{2k+1} = \dim V[1] - \dim V[2l+3] = \dim V[1]$ (since $2l+2, 2l+3 \geq m$). So, there's no ambiguity to conclude that $\sum n_{2k} = \dim V[0]$, and $\sum n_{2k+1} = \dim V[1]$.

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Problem 4

Show that the symmetric power representation $S^k\mathbb{C}^2$ is isomorphic to the irreducible representation V_k with the highest weight k.

Solution: First, fix basis $x, y \in \mathbb{C}^2$. Since $\mathfrak{sl}(2,\mathbb{C}) \subset \mathfrak{gl}(2,\mathbb{C}) = \mathrm{End}(\mathbb{C}^2)$, there is a natural $\mathfrak{sl}(2,\mathbb{C})$ -action on $\mathbb{C}^2$, by simply viewing it as a subspace of the linear operators on $\mathbb{C}^2$.

I. Map of Tensor Representation:

Before starting, we want to understand the action of $\mathfrak{sl}(2,\mathbb{C})$ on $(\mathbb{C}^2)^{\otimes n}$ first (and we'll do so by applying induction): Recall that given $\rho_V : \mathfrak{g} \to \mathfrak{gl}(V)$ and $\rho_W : \mathfrak{g} \to \mathfrak{gl}(W)$ two Lie Algebra Representation, the Tensor Representation $\rho_{V\otimes W} : \mathfrak{g} \to \mathfrak{gl}(V \otimes W)$ is defined by $\rho_{V\otimes W}(x) = \rho_V(x) \otimes 1_W + 1_V \otimes \rho_W(x)$.

For base case $n = 2$, a Lie Algebra Representation $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$, the Tensor Representation on $V \otimes V$ is defined as $\rho_2 : \mathfrak{g} \to \mathfrak{gl}(V \otimes V)$ by $\rho_2(x) := \rho(x) \otimes 1_V + 1_V \otimes \rho(x)$. Now, suppose for given $n \in \mathbb{N}$, the Tensor Representation $\rho_n : \mathfrak{g} \to \mathfrak{gl}(V^{\otimes n})$ is given as follow:

$$\rho_n(x) = \sum_{i=1}^{n} x_i^{(n)}, \qquad x_i^{(n)} := 1_V \otimes ... \otimes \rho(x) \otimes ... \otimes 1_V,\ \ \rho(x) \text{ at } i^{\text{th}} \text{ position} \tag{4.1}$$

Then, the corresponding Tensor Representation $\rho_{n+1} : \mathfrak{g} \to \mathfrak{gl}(V^{\otimes n+1})$ (where $V^{\otimes n+1} = V^{\otimes n} \otimes V$) is given as follow:

$$\rho_{n+1}(x) = \rho_n(x) \otimes 1_V + 1_{V^{\otimes n}} \otimes \rho(x) = \left(\sum_{i=1}^{n} x_i^{(n)}\right) \otimes 1_V + 1_V \otimes ... \otimes 1_V \otimes \rho(x) \tag{4.2}$$

Where, the second tensor $1_V \otimes ... \otimes 1_V \otimes \rho(x)$ has n copies of 1_V with $\rho(x)$ at the $(n+1)^{\text{th}}$ position, we'll denote as $x_{n+1}^{(n+1)}$. Also, each $x_i^{(n)} = 1_V \otimes ... \otimes \rho(x) \otimes ... \otimes 1_V$ (total of n entries, with $\rho(x)$ at the i^{th} entry), then $x_i^{(n)} \otimes 1_V$ has $(n+1)$ entries, with $\rho(x)$ at the i^{th} entry, hence $x_i^{(n)} \otimes 1_V = x_i^{(n+1)}$. So, eventually the above simplivies to:

$$\begin{aligned} \rho_{n+1}(x) &= \left(\sum_{i=1}^{n} x_i^{(n)}\right) \otimes 1_V + x_{n+1}^{(n+1)} \\ &= \sum_{i=1}^{n} x_i^{(n)} \otimes 1_V + x_{n+1}^{(n+1)} = \sum_{i=1}^{n} x_i^{(n+1)} + x_{n+1}^{(n+1)} = \sum_{i=1}^{n+1} x_i^{(n+1)} \end{aligned} \tag{4.3}$$

Which is the desired form of the tensor representation. So, based on induction, the n-Tensor Representation $\rho_n : \mathfrak{g} \to \mathfrak{gl}(V^{\otimes n})$ is given by $\rho_n(x) = \sum_{i=1}^{n} 1_V \otimes ... \otimes \rho(x) \otimes ... \otimes 1_V$ (where for each index i, $\rho(x)$ is at the i^{th} entry).

II. Symmetric Tensor of $\mathbb{C}^2$:

Given $V^{\otimes k}$, the symmetric tensor S^kV can be defined as the quotient $V^{\otimes k}/S_k$, where we say $\otimes_{i=1}^k v_i \sim \otimes_{i=1}^k u_i$, if there exists a permutation $\sigma \in S_k$, such that each $u_i = v_{\sigma(i)}$ (i.e. identify each pure k-tensor to be the same, if the entries are the same up to coordinates permutation).

Which, if limit $V = \mathbb{C}^2$, with basis x, y, then$S^k\mathbb{C}^2$ in fact has a basis of the collection $\left\{v_{pq} := (\otimes_{i=1}^p x) \otimes \left(\otimes_{j=1}^q y\right) \mid p + q = k\right\}$ (since $(\mathbb{C}^2)^{\otimes k}$ is spanned by all the pure k-tensors formed with various permutations of x, y with total of k entries, then since here the equivalence is up to permutation of entries, $S^k\mathbb{C}^2$ is spanned by these pure k-tensors that covered all possible amount of x, y that fit as a k-tensor; the reason they're linearly independent, is due to the fact that if the pair (p, q) don't match, the corresponding v_{pq} are not equivalent under S_k-action, so no linaer combinations of them would ever be quotiented out).

This shows that $\dim(S^k\mathbb{C}^2) = k+1$, which as vector space $S^k\mathbb{C}^2 \cong V_k$ (since $V_k \subset \mathbb{C}[x,y]$ is spanned by all $x^p y^q$ where $p+q=k$, so $\dim(V_k) = k+1$ also). Here, an explicit isomorphism is given by $T : S^k\mathbb{C}^2 \xrightarrow{\sim} V_k$ by $T\Big(\big(\otimes_{i=1}^p x\big) \otimes \big(\otimes_{j=1}^q y\big)\Big) = x^p y^q$.

III. Invariance of $\mathfrak{sl}(2,\mathbb{C})$ action on $S^k\mathbb{C}^2$:

Given any tensor $\otimes_{i=1}^k v_i \in (\mathbb{C}^2)^{\otimes k}$, given any $x \in \mathfrak{sl}(2,\mathbb{C})$, the tensor representation is given as:

$$\rho_k(x) \cdot \big(\otimes_{i=1}^k v_i\big) = \sum_{i=1}^k x_i^{(k)} \cdot \big(\otimes_{i=1}^k v_i\big), \quad x_i^{(k)} \cdot \big(\otimes_{i=1}^k v_i\big) = v_1 \otimes \ldots \otimes (x \cdot v_i) \otimes \ldots \otimes v_k \tag{4.4}$$

So, chosen any permutation $\sigma \in S_k$, we have the following:

$$\rho_k(x) \cdot \big(\otimes_{i=1}^k v_{\sigma(i)}\big) = \sum_{i=1}^k x_i^{(n)} \cdot \big(\otimes_{i=1}^k v_{\sigma(i)}\big) = \sum_{i=1}^k x_{\sigma(i)}^{(k)} \cdot \big(\otimes_{i=1}^k v_i\big) = \rho_k(x) \cdot \big(\otimes_{i=1}^k v_i\big) \tag{4.5}$$

(Note: since $\big(\otimes_{i=1}^k v_i\big)$ is fixed, one can permute $x_i^{(k)}$ instead fo the indices to match up). Hence, this shows that the $\mathfrak{sl}(2,\mathbb{C})$ action on $(\mathbb{C}^2)^{\otimes k}$ is invariant under S_k-action, which implies that the $\mathfrak{sl}(2,\mathbb{C})$-action on $S^k\mathbb{C}^2 := (\mathbb{C}^2)^{\otimes k}/S_k$ is well-defined, and acts the same on their representatives in $(\mathbb{C}^2)^{\otimes k}$. So, for ease of calculation, we'll use the representatives in $(\mathbb{C}^2)^{\otimes k}$ to calculate the action, then take quotient to see the result in $S^k\mathbb{C}^2$.

III. Isomorphism as $\mathfrak{sl}(2,\mathbb{C})$ Representation:

Finally, we aim to show that $T : S^k\mathbb{C}^2 \xrightarrow{\sim} V_k$ in part **II** is in fact an isomorphism as $\mathfrak{sl}(2,\mathbb{C})$ representations. Since it's already a linear isomorphism, it suffices to check if T is a morphism of $\mathfrak{sl}(2,\mathbb{C})$ representations.

Given the matrix form $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, the action on $\mathbb{C}^2$ (with ordered basis $\{x, y\}$) is as follow:

$$h \cdot x = x, \quad h \cdot y = -y, \qquad e \cdot x = 0, \quad e \cdot y = x, \qquad f \cdot x = y, \quad f \cdot y = 0 \tag{4.6}$$

Hence, given any $v_{pq} = \big(\otimes_{i=1}^p x\big) \otimes \big(\otimes_{j=1}^q y\big)$, the action of h, e, f on $S^k\mathbb{C}^2$ are as follow:

- For h, each $h_i^{(k)} = 1_V \otimes \ldots \otimes h \otimes \ldots \otimes 1_V$ (with h at the i^{th} entry, identity on the rest). Hence, if $p < i$, the i^{th} entry of v_{pq} here is given by y, so $h_i^{(k)} v_{pq} = -v_{pq}$ (since at the i^{th} entry $h \cdot y = -y$); else, if $p \geq i$, the i^{th} entry of v_{pq} is given by x, so $h_i^{(k)} v_{pq} = v_{pq}$ (since $h \cdot x = x$ on the i^{th} entry). Therefore:

$$\begin{aligned} \rho_k(h) \cdot v_{pq} &= \sum_{i=1}^k h_i^{(k)} \cdot v_{pq} = \sum_{i=1}^p h_i^{(k)} \cdot v_{pq} + \sum_{i=p+1}^k h_i^{(k)} \cdot v_{pq} \\ &= p \cdot v_{pq} - (k-p) \cdot v_{pq} = (p-q) \cdot v_{pq} \end{aligned} \tag{4.7}$$

  (Note: Recall that $p + q = k$).

- For e, each $e_i^{(k)}$ has e at the i^{th} entry (identity on the rest). Hence, if $p < i$, the i^{th} entry of v_{pq} is given by y, so $e_i^{(n)} v_{pq} \sim v_{(p+1)(q-1)}$ (since on the i^{th} entry we have $e \cdot y = x$, so increase number of x i.e. p by 1, and decrease number of y i.e. q by 1); else if $p \geq i$, the i^{th} entry of v_{pq} is given by x, so $e_i^{(n)} v_{pq} = 0$ (since i^{th} entry is $e \cdot x = 0$). Hence:

$$\rho_k(e)\cdot v_{pq} = \sum_{i=1}^{k} e_i^{(k)}\cdot v_{pq} = \sum_{i=1}^{p} e_i^{(k)}\cdot v_{pq} + \sum_{i=p+1}^{k} e_i^{(k)}\cdot v_{pq} \\ = (k-p)v_{(p+1)(q-1)} = q\cdot v_{(p+1)(q-1)} \tag{4.8}$$

- For f, each $f^{(k)_i}$ has f at the i^{th} entry (identity on the rest). Hence, if $p < i$, the i^{th} entry of v_{pq} is given by y, so $f^{(n)_i v_{pq}} = 0$ (since $f\cdot y = 0$ on the i^{th} entry); else, if $p \geq i$, the i^{th} entry of v_{pq} is x instead, so $f_i^{(n)}v_{pq} \sim v_{(p-1)(q+1)}$ (since $f\cdot x = y$ on the i^{th} entry, so number of x i.e. p decrease by 1, while number of y i.e. q increase by 1). Hence:

$$\rho_k(f)\cdot v_{pq} = \sum_{i=1}^{k} f_i^{(k)}\cdot v_{pq} = \sum_{i=1}^{p} f_i^{(k)}\cdot v_{pq} + \sum_{i=p+1}^{k} f_i^{(k)}\cdot v_{pq} = p\cdot v_{(p-1)(q+1)} \tag{4.9}$$

So, with the map T in mind, the $\mathfrak{sl}(2,\mathbb{C})$-action have h, e, f satisfy the following:

$$T\big(\rho_k(h)\cdot v_{pq}\big) = T\big((p-q)\cdot v_{pq}\big) = (p-q)x^p y^q = h\cdot(x^p y^q) = h\cdot T\big(v_{pq}\big) \tag{4.10}$$

$$T\big(\rho_k(e)\cdot v_{pq}\big) = T\Big(q\cdot v_{(p+1)(q+1)}\Big) = qx^{p+1}y^{q-1} = e\cdot(x^p y^q) = e\cdot T\big(v_{pq}\big) \tag{4.11}$$

$$T\big(\rho_k(f)\cdot v_{pq}\big) = T\Big(p\cdot v_{(p-1)(q+1)}\Big) = px^{p-1}y^{q-1} = f\cdot(x^p y^q) = f\cdot T\big(v_{pq}\big) \tag{4.12}$$

(Note: above h, e, f acts on the monomials $x^p y^q$ as differential operators).

Hence, we concluded that T is also a morphism of $\mathfrak{sl}(2,\mathbb{C})$ Representation (since it preserves the action of h, e, f on the vector spaces, which are the basis elements of $\mathfrak{sl}(2,\mathbb{C})$). Therefore, $S^k\mathbb{C}^2 \cong V_k$ not only as vector spaces, but also as $\mathfrak{sl}(2,\mathbb{C})$-Representations.